

Lily Bernard at Learngame-HC College

Context

Ludo Bernard is the principal at Learngame-HC College and wishes the best results for his daughter, Lily Bernard, a final year student at the college. Here, all that matters is the z-score, one of the most common measures in statistics to gauge and compare performance among students. It's the z-score that counts for university admissions.

For a subject, if n is the number of students and $x_1, x_2 \dots x_n$ are the final grades of the students, the average and standard deviation of the cohort are calculated as follows:

$$\begin{aligned}\mu &= \frac{1}{n} \sum_{i=1}^n x_i \\ \sigma &= \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \\ &= \sqrt{\frac{\sum_{i=1}^n x_i^2}{n} - \mu^2}\end{aligned}$$

Then the z-score of a student A is calculated as follows:

$$z_A = \frac{x_A - \mu}{\sigma}$$

In the mathematics course, the final grades of all students have been recorded. Lily has a good grade, so her z-score is also very good. However, since she wants to study medicine, her father Ludo is worried and wonders if her z-score will be sufficient.

Being the principal, he decides to manipulate the students' grades to improve Lily's z-score. However, to avoid suspicion, he must not change the grades randomly. He sets 4 rules to avoid being exposed:

1. **Do not change Lily's grade.** Indeed, as she is his daughter (conflict of interest), her grade may be checked by a third party.
2. **Do not lower the grade of any student.** Indeed, a student surprised negatively by his grade is more likely to request a review, which would expose the scheme.
3. **If a student A has a higher grade than a student B before cheating, then student A must have a grade greater than or equal to student B after cheating.** If two friends A and B who talk a lot think that A did better in the evaluations, but B ends up with a better grade, they may suspect foul play.
4. **Grades must remain multiples of 0.01%.** Indeed, the format of the grades does not allow for precision greater than one hundredth of a percentage.

In short, the principal allows himself to increase the grades of any subset of students, as long as the order of the grades is preserved and his daughter's grade remains unchanged.

Your Mission

The file `data.txt` contains the blank grades of the students. Your mission is to modify the grades while respecting the previous 4 constraints, in order to optimize Lily's z-score.

The answer to the puzzle is a string in the same format as the one in `data.txt`, but for which the grades have been modified.

Flag

The file `'flag-steps-fake.txt'` serves as an example to guide you on how to create the FLAG. Except for one big detail, reproduce exactly the steps in `'flag-steps-fake.txt'`.

The big detail is obviously that you have to have to change the grades of the students before applying these steps.